CLIMB

When to Call the Team

Save tokens on multi-agent LLM pipelines, with a guarantee on accuracy

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EXECUTIVE SUMMARY

Call the team only when you must, with a guarantee

Problem	Agent teams are more accurate, but use up to **6.3×** the tokens of a single agent and break right answers on up to **9%** of tasks.
Idea	Try cheap answers first. Call the team only when a cheap answer looks unreliable (**confidence**, **agreement**).
Guarantee	You set the accuracy you can lose. The **data sets the threshold**: a **betting test** certifies the cheapest one that keeps it, **90%** of the time.
Result	**38–65% fewer tokens** on 3 models × 4 benchmarks. The promise broke in **≤ 2%** of splits (10% allowed).
For Amex	Sits in front of any multi-agent workflow. Lessons for designing complex agent systems.

BACKGROUND

What is an agent team?

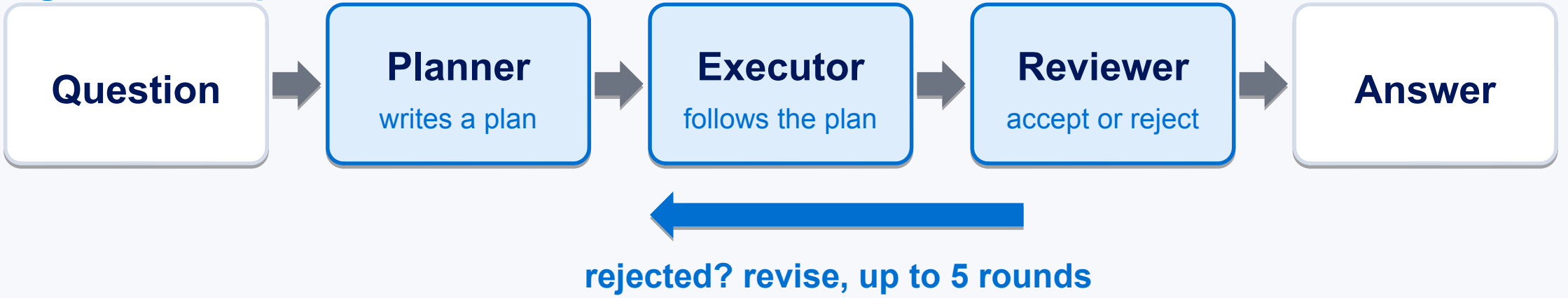
Single agent



+2 to +21

points over a direct answer. Best on 12 / 12 settings.

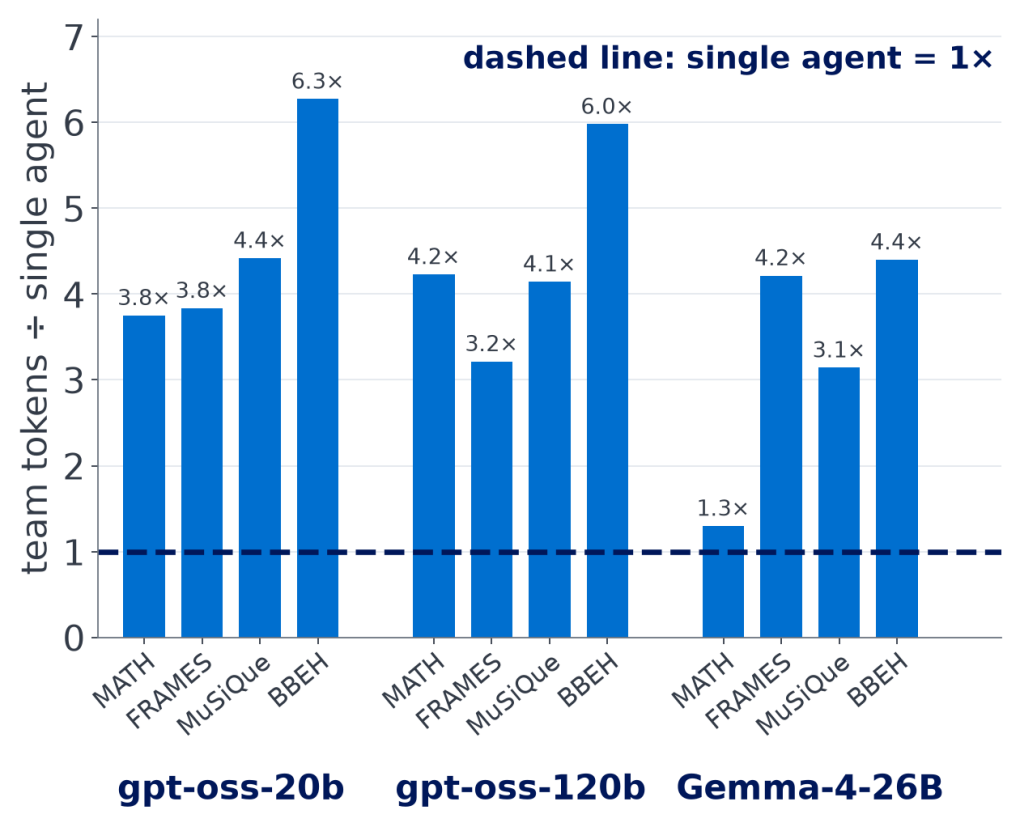
Agent team: plan, execute, review



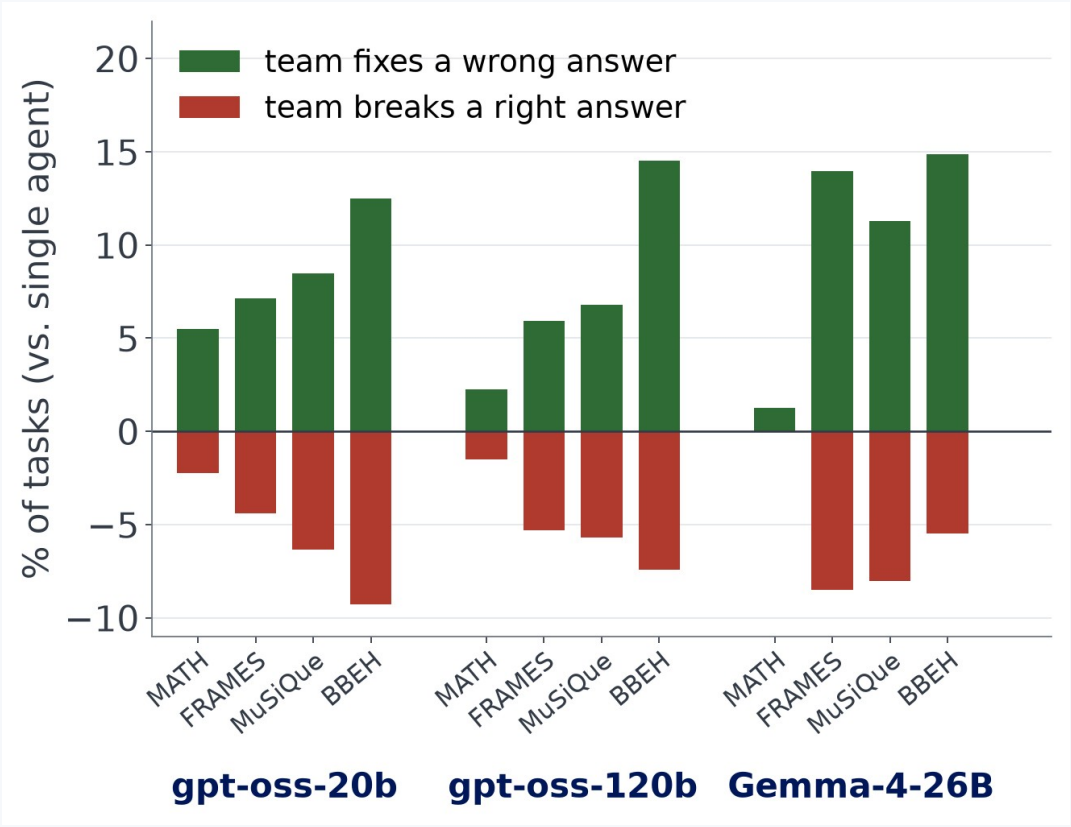
THE CATCH

Teams are costly, and sometimes harmful

Cost: up to 6.3× a single agent



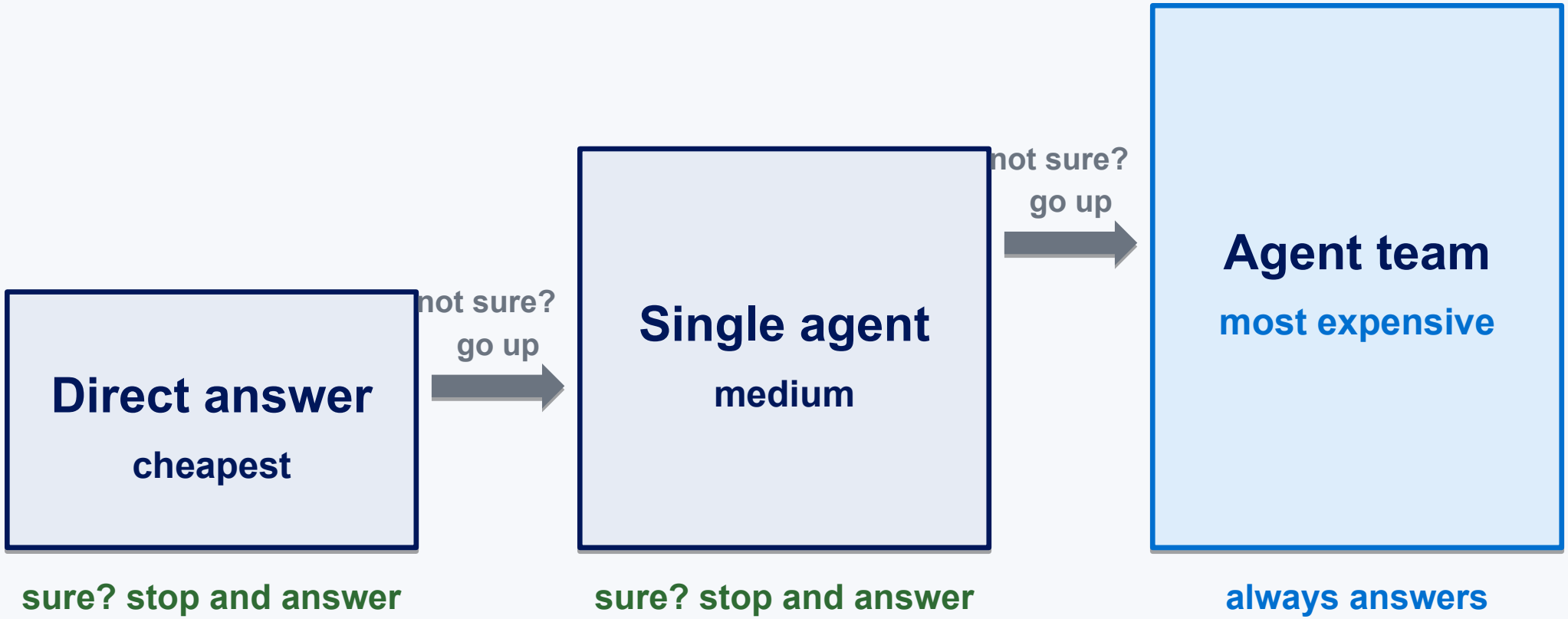
Harm: up to 9% of tasks broken



THE IDEA

Climb a ladder: call the team only when needed

Stop if score > threshold. The key question: which threshold?



STEP 1 · THE SCORE

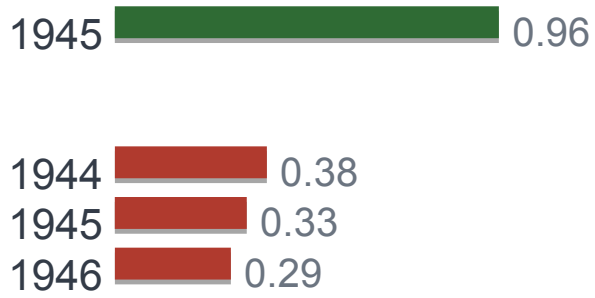
Why confidence, and why agreement?

A good signal tells right from wrong, and costs far less than the team.

Confidence

Is the model sure of its answer?

Example: “When did World War II end?”



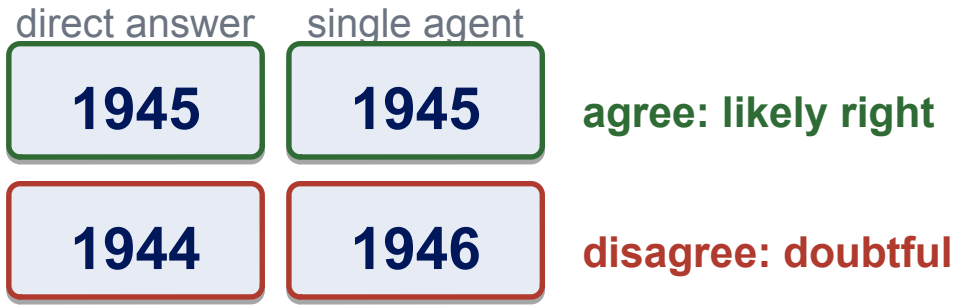
A model that knows tends to put its probability on **one** answer. An unsure one spreads it.

Cost: free, it comes with the answer.

Agreement

Do two cheap answers match?

Example: “When did World War II end?”



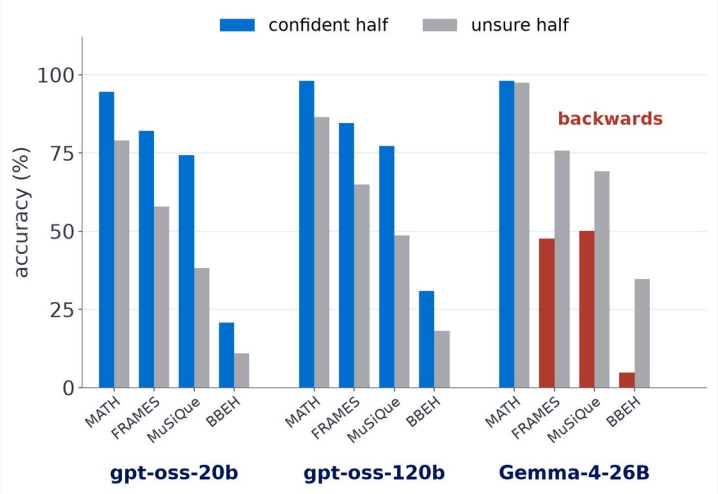
One right answer, many wrong ones: two tries **rarely** match on a wrong one.

Cost: one more cheap call, no log-probs.

STEP 1 · THE SCORE

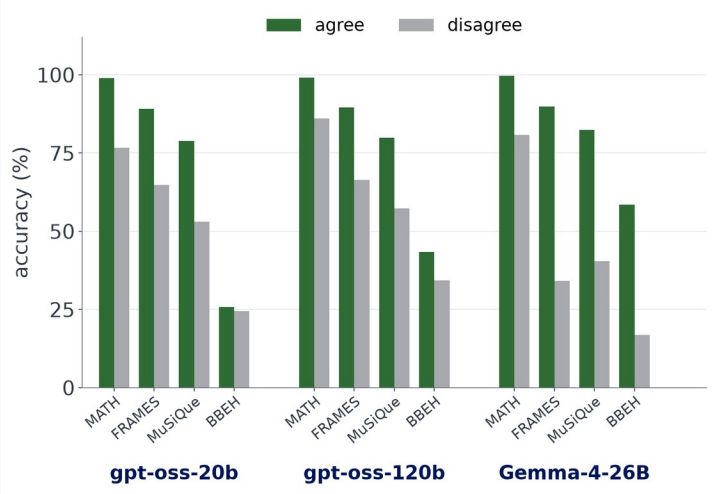
Do they work? Mostly, and they cover each other

Confidence: direct answer accuracy



Mostly helps. **Backwards on 3 (Gemma).**

Agreement: single agent accuracy



Mostly helps, **including those 3.**

When confidence is backwards, CLIMB still holds:

1 Agreement first

Option B ranks by agreement. Confidence only breaks ties.

2 The check catches it

A backwards score fails the test early. It is not deployed.

3 Worst case: the team

Less saving, no lost accuracy. Gemma: 69–96% of tokens.

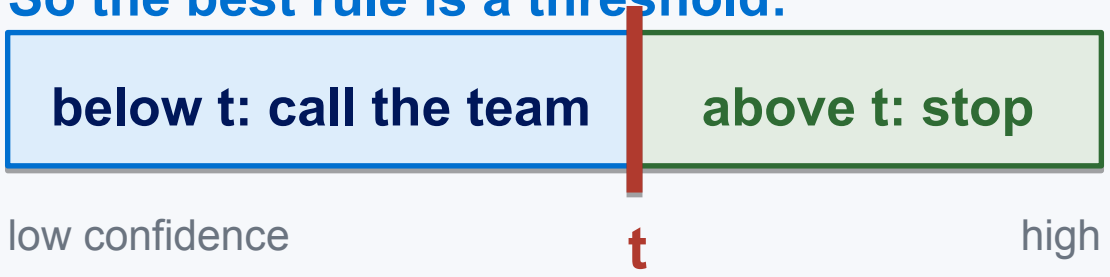
STEP 2 · SETTING THE THRESHOLD

Why a threshold? It is the best rule

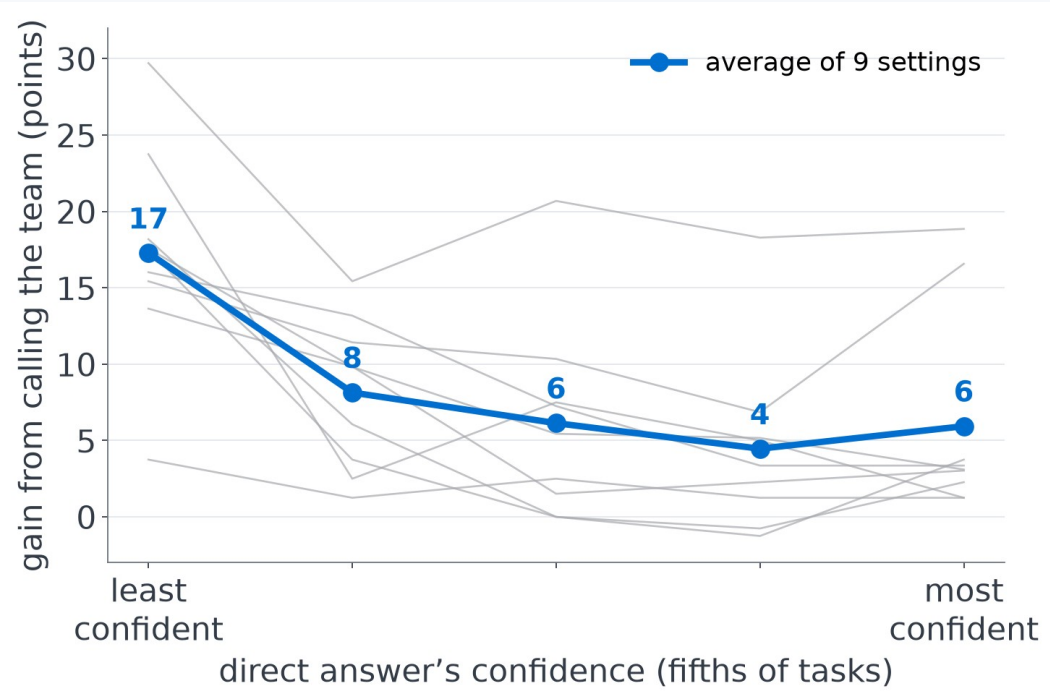
Stop the tasks where the team adds **least**.

The team adds least where confidence is **high**.

So the best rule is a threshold:



Gain from the team, by confidence

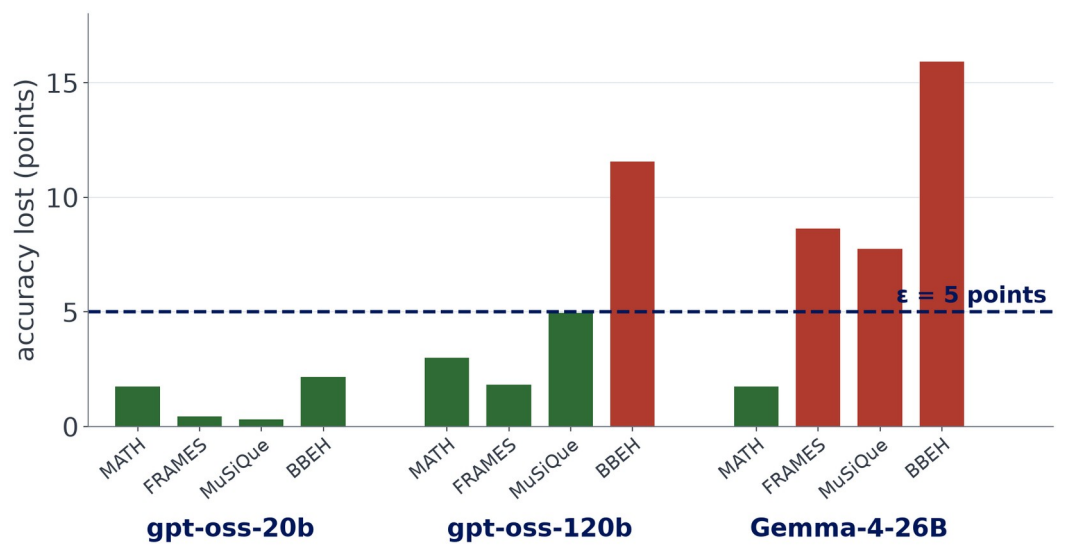


Theory gives the **form**. **The data decides where t goes.**

STEP 2 · SETTING THE THRESHOLD

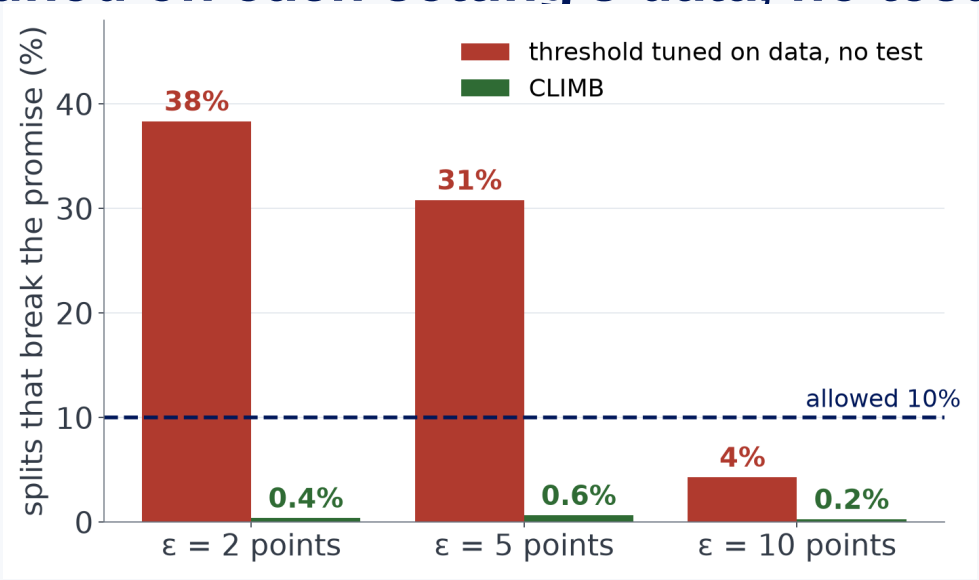
Two shortcuts, and both break the promise

One threshold for all settings



Above ε on **4 of 12 settings, up to 16 points.**

Tuned on each setting's data, no test



Breaks the promise on **31% of splits** at ε = 5.

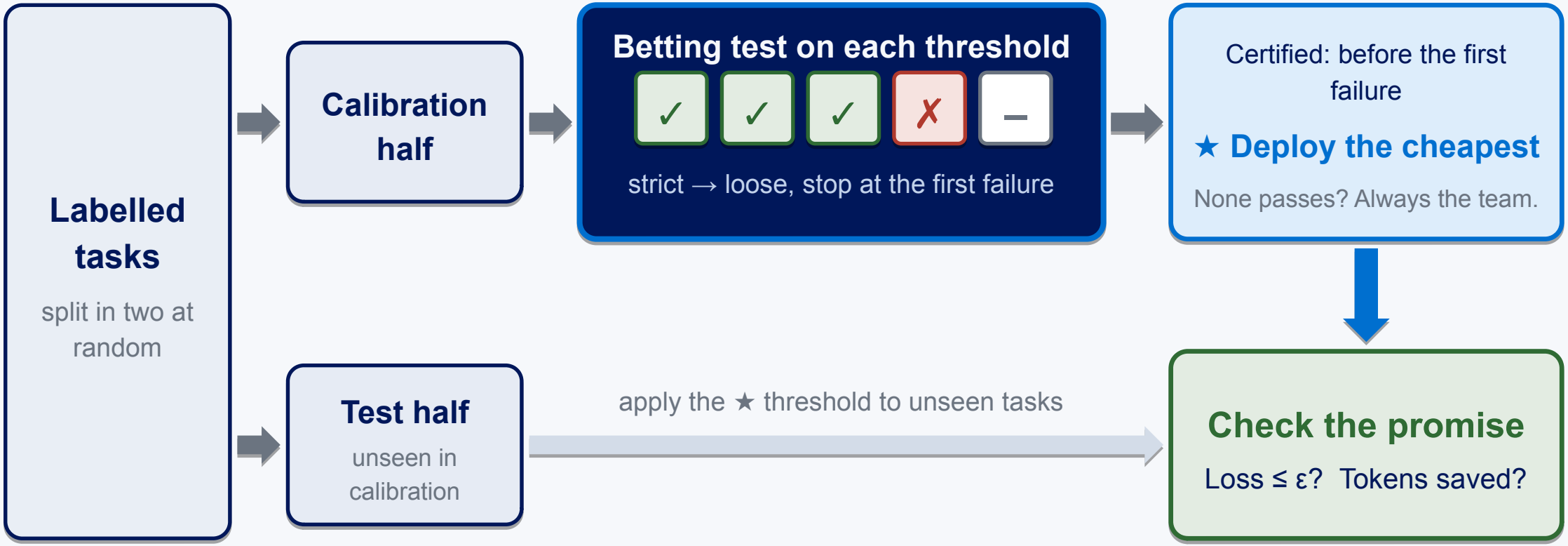
The right threshold depends on the model and task.

We need it set from each setting's data, with a guarantee.

STEP 2 · SETTING THE THRESHOLD

Replace guessing with a promise: calibrate, then test

You pick only ε, e.g. “lose at most 5 points”. The data picks the threshold.



In our experiments: 200 random splits. In deployment: only the calibration half, the test half is your future tasks.

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

Why a betting test?

We need: accept an unsafe threshold at most 5% of the time, from a few hundred tasks.

Saves count

A cheap answer can also save one the team breaks (up to 9%). The bet credits it.

Uses low variance

Most tasks lose 0. The bet size adapts to what it has seen.

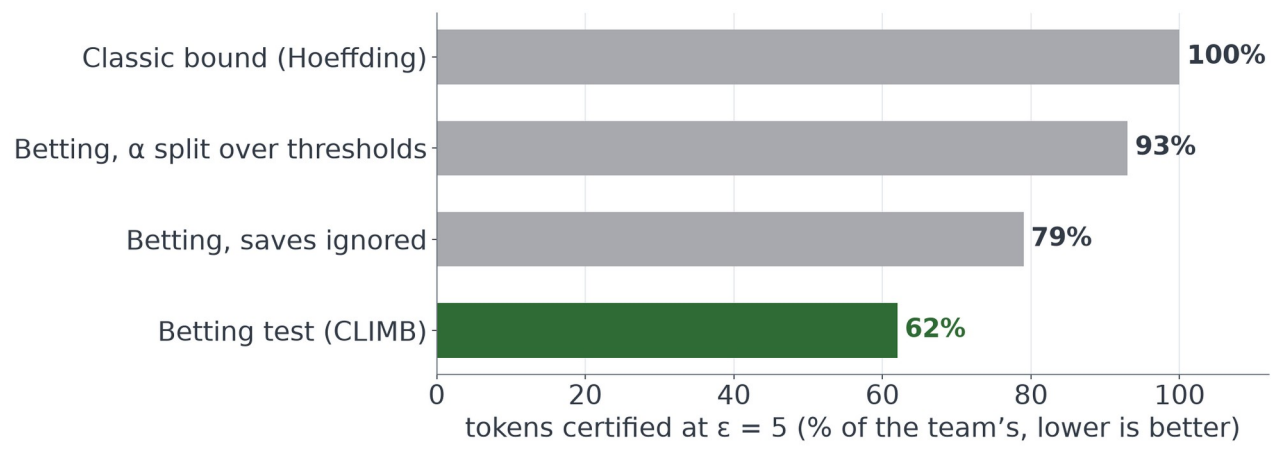
Many thresholds, full 5%

Tested strict → loose, each test keeps the whole 5%.

Any data

No model of the score. Only: calibration looks like deployment.

All valid. Which one saves the most?



62% vs 79–100%

of the team's tokens at ε = 5

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

The betting test: start with $1, bet the threshold is safe

Reveal the labelled tasks one by one. Each task multiplies the money:

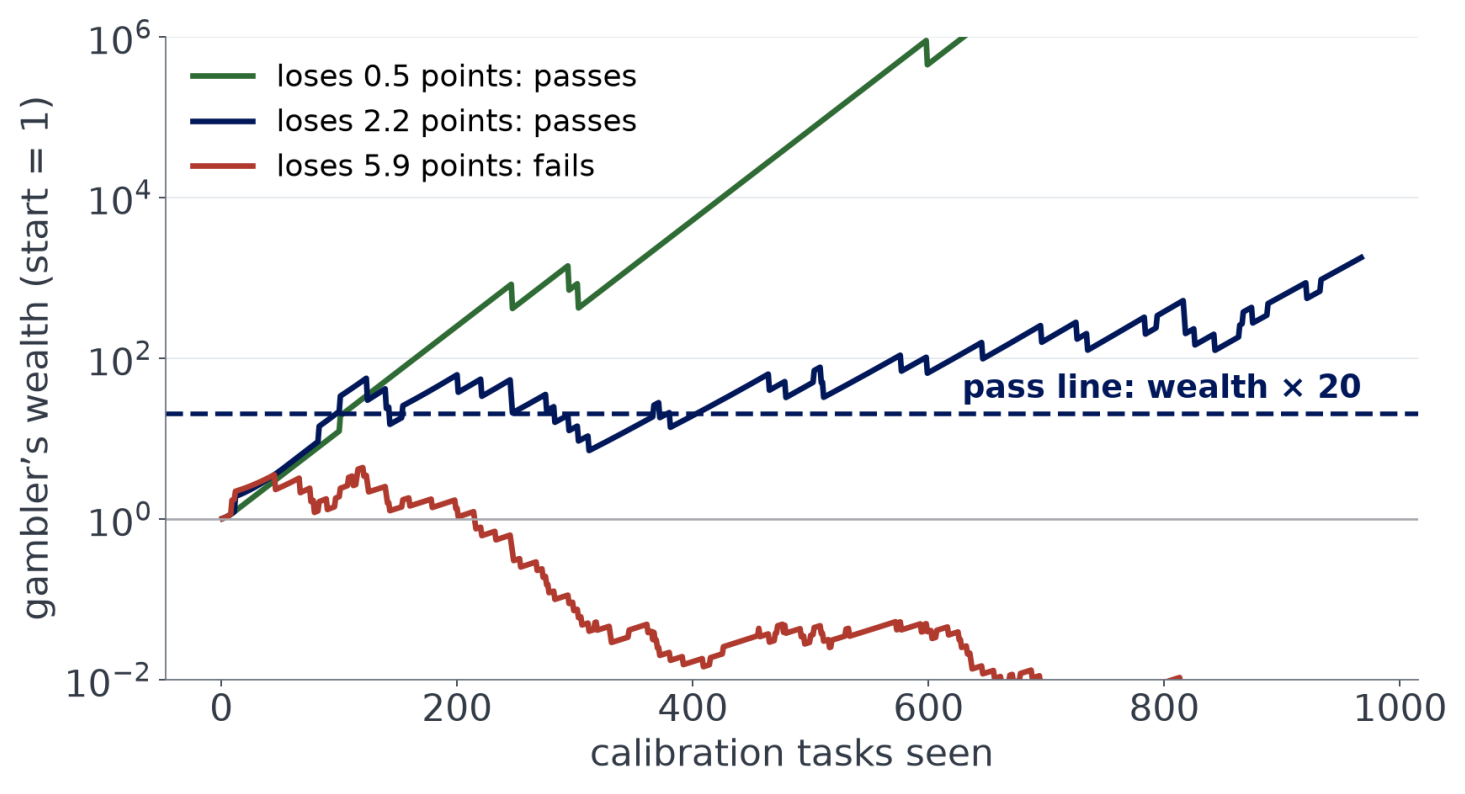
Cheap wrong, team right	Same result	Cheap right, team wrong
× 0.53	× 1.03	× 1.53
lose a lot	win a little (most tasks)	win a lot

Money ever reaches $20 → pass. Never reaches it → fail.

Multiplier = 1 + λ (ε − loss), loss = 1, 0 or −1. Shown for ε = 5 points and bet size λ = 0.5.

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

Safe thresholds get rich, unsafe ones go broke



Average multiplier

$$1 + \lambda(\varepsilon - \mu)$$

μ: the threshold's true average loss

Safe: μ < ε

average > 1: money grows, passes

Unsafe: μ > ε

average < 1: money shrinks, fails

Bet size λ: set from earlier tasks only.

Why $20? An unsafe threshold rarely gets there

Unsafe threshold
each bet loses on average

→

Average ≤ $1
never grows on average

→

Never below $0
a loss at most halves it

$$1 \geq \text{average wealth} \geq 20 \times P(\text{reach \$20})$$

so P(an unsafe threshold passes) ≤ 1/20 = 5%

From one test to a threshold we can deploy

One threshold
An unsafe one passes ≤ 5%.

→

Strict → loose
Stop at the first failure. An unsafe one gets in only if the **first** unsafe one passes: ≤ 5%.

→

Certified set
All safe with prob. ≥ 95%.
★ Deploy the cheapest.

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

In CLIMB: testing one threshold, task by task

Task	Score > t?	Our answer	Team	Loss ℓ	× money	Money
1	yes, stop	✓	✓	**0**	× 1.03	**$1.03**
2	no → team	team's	✓	**0**	× 1.03	**$1.05**
3	yes, stop	✗	✓	**+1 lost**	× 0.53	**$0.55**
4	yes, stop	✓	✗	**−1 saved**	× 1.53	**$0.84**
5	yes, stop	✓	✓	**0**	× 1.03	**$0.86**
…						**≥ $20?**

ε = 5 points, bet size λ = 0.5. Escalated tasks use the team's answer: loss 0.

Each threshold, strict → loose

1 Tasks in random order, $1.

2 ℓ = team right − ours right.

3 λ from **earlier tasks only**. A loss at most halves the money.

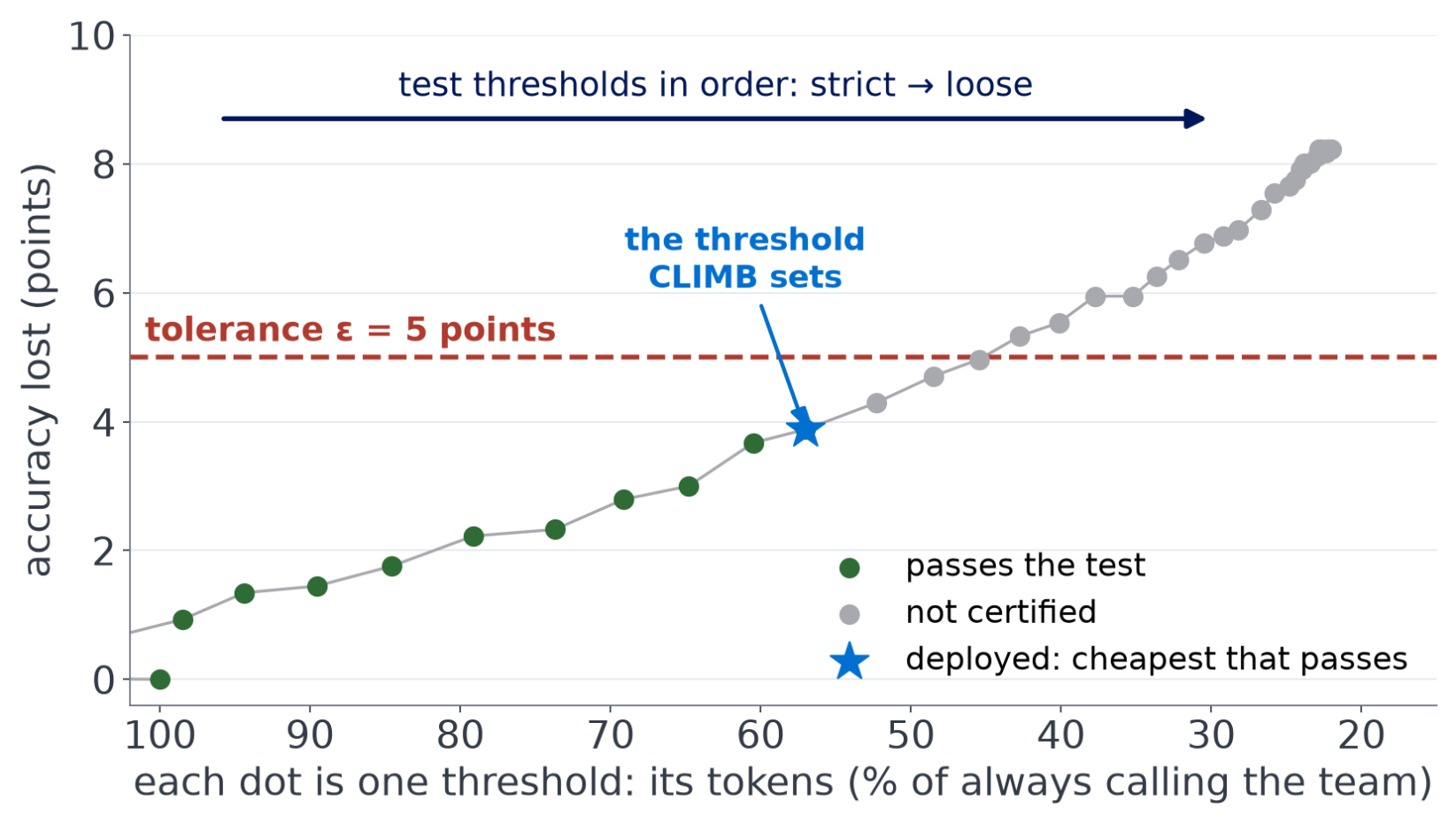
4 Money × (1 + λ (ε − ℓ)).

5 Ever ≥ $20: certified, next threshold. Never: stop.

Options A and B are each tested this way at 5%, so both hold together with probability ≥ 90%.

STEP 2 · SETTING THE THRESHOLD

Putting it together: how CLIMB sets the threshold

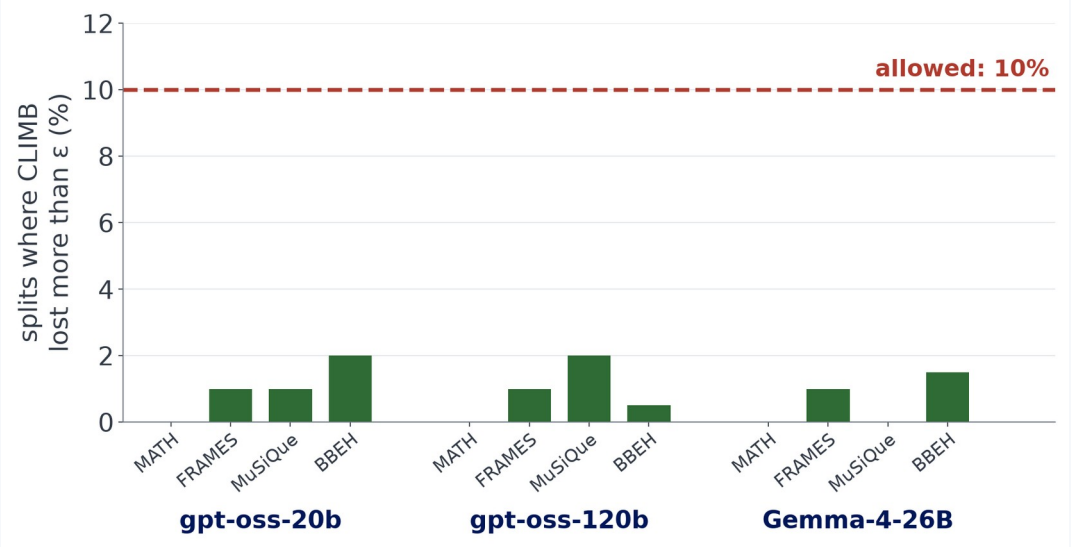


gpt-oss-120b on MuSiQue, threshold on the direct answer's confidence, one calibration half. Two-rung thresholds are tested the same way, the cheaper is kept.

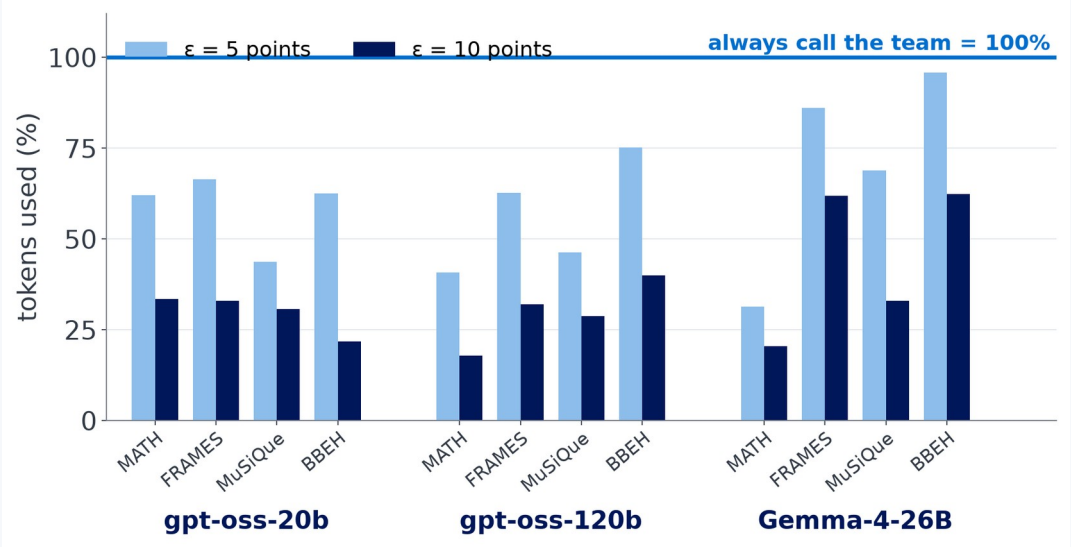
RESULTS · 3 MODELS × 4 BENCHMARKS, 200 RANDOM SPLITS

These thresholds keep the promise and save 38–65%

Splits that broke the promise (%)



Tokens used (% of the team's)



≤ 2%

of splits broke the promise. Allowed: 10%.

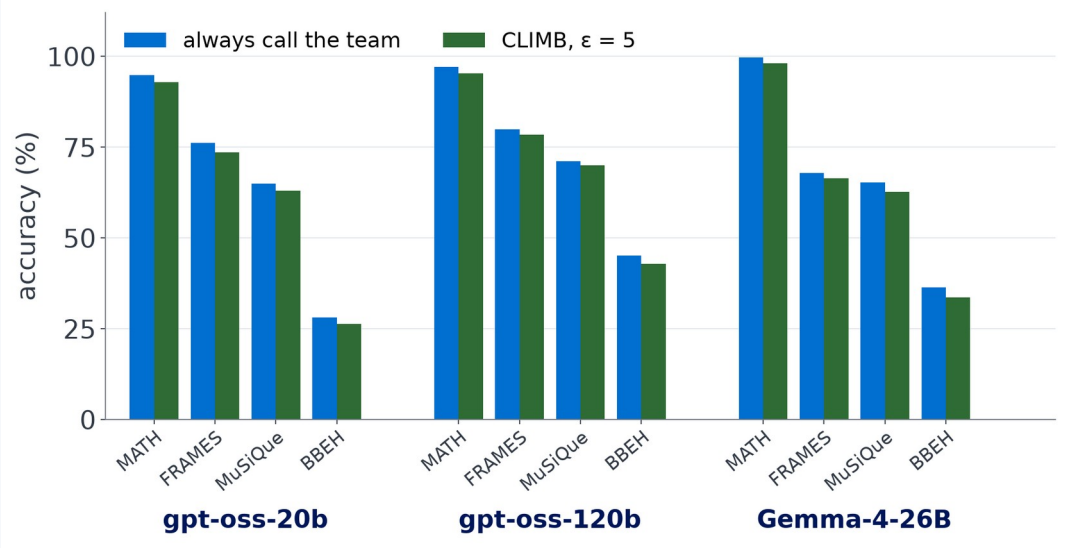
38% / 65% saved

allowing 5 / 10 points of loss

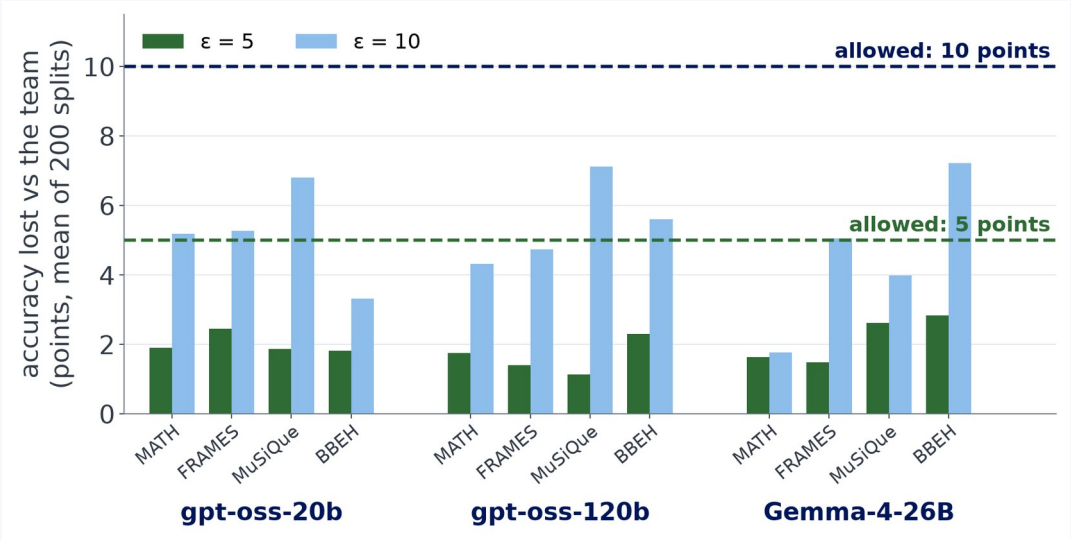
RESULTS · 3 MODELS × 4 BENCHMARKS, 200 RANDOM SPLITS

Accuracy with CLIMB stays within the loss you allow

Accuracy (%)



Points lost vs always calling the team



1.1–2.8 points lost

allowed 5, on average over splits

1.8–7.2 points lost

allowed 10, on average over splits

TAKEAWAYS

Call the team only when you must

Teams are costly

and sometimes break right answers.

Data-set threshold

you set the loss, a betting test finds the cheapest safe threshold.

38–65% saved

at 5 and 10 points, with the promise kept.

Assumes calibration tasks resemble deployment tasks. Recalibrating takes under a second.

FOR AMEX

What this offers Amex

1 Fewer tokens, even on closed models

Closed model writes the output → **Open model** reads it → log-probs → **CLIMB** uses them as S_0

Same idea as the reference model in PPO.

- Feed the closed model's output to an open model to get its **log-probs**
- **38–65% fewer tokens** at 5–10 points in our tests
- Agreement of two cheap calls needs **no log-probs**
- A weaker score saves less, the **promise still holds**

Getting a closed model's log-probs from an open model is a proposal, not yet tested.

2 Lessons for complex agent systems

More agents is not always better
Reviewers pass 23–58% of wrong drafts.

Route each task, not the whole system
Escalate only the doubtful tasks.

Cheap signals are strong
Two cheap answers agree → usually right.

Put a guarantee on any routing rule
Cascades, tool calls, human review.

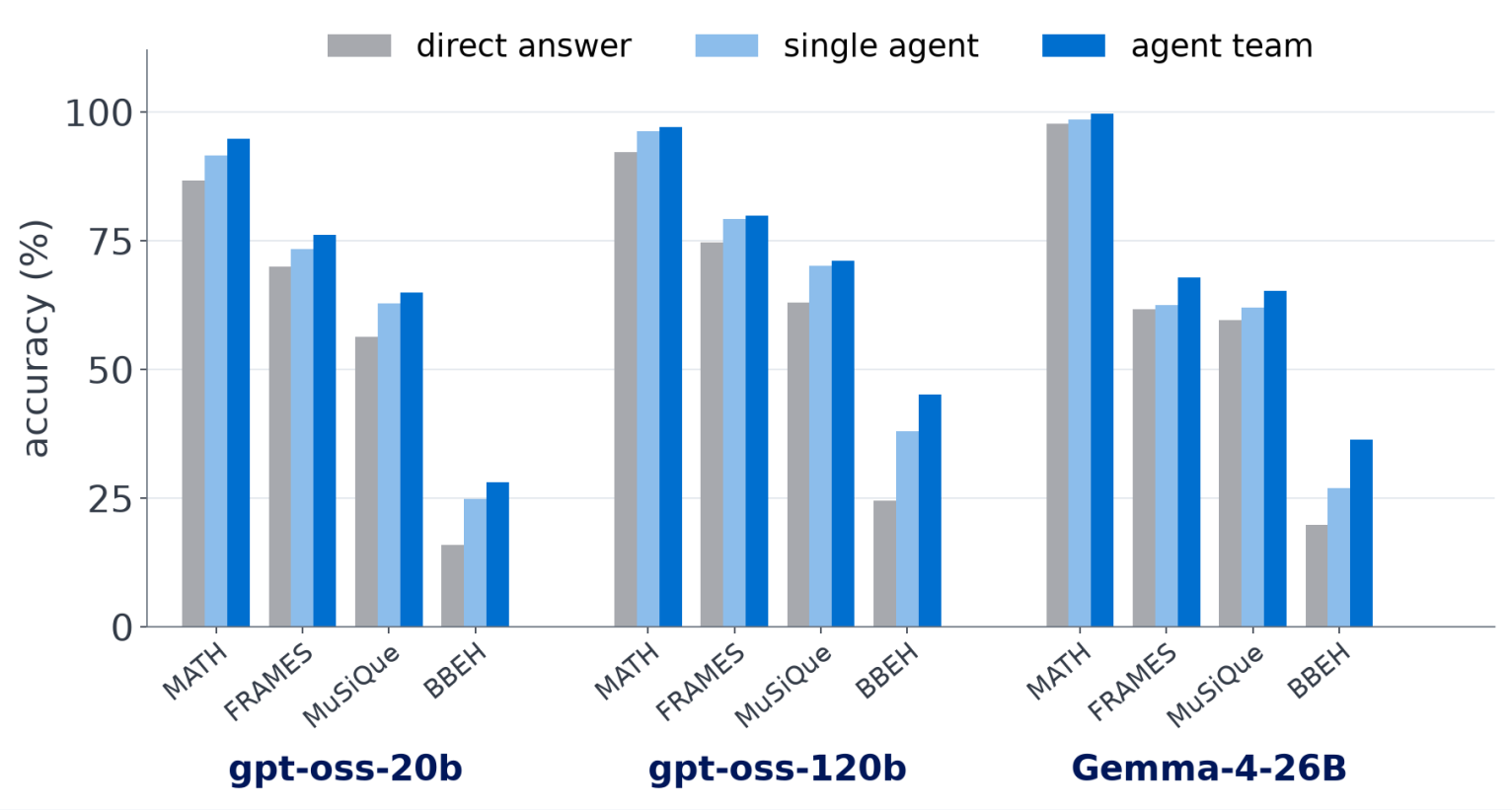
Thank you. Questions?

Backup

Details for questions

BACKGROUND

Teams are more accurate



+2 to +21

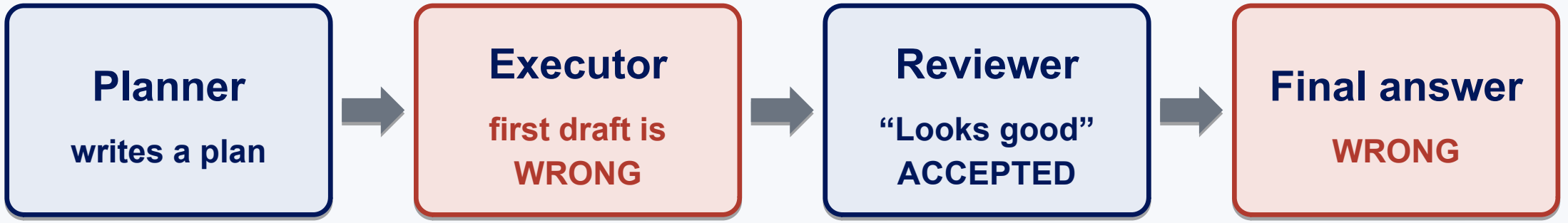
points over a direct answer

12 / 12

settings where the team is best, up to 9 points over a single agent

THE PROBLEM

Why does the team break right answers?



84%

of broken answers start as a wrong first draft the reviewer did not catch

23–58%

of wrong answers are accepted by the reviewer

HOW TO CLIMB

Deciding when to climb: four questions

1

Signal

Does the cheap answer look right?

→ **confidence and agreement**

2

Rule

Where do we draw the line?

→ **a threshold on the score**

3

Check

Is that line safe?

→ **a betting test: the core**

4

Depth

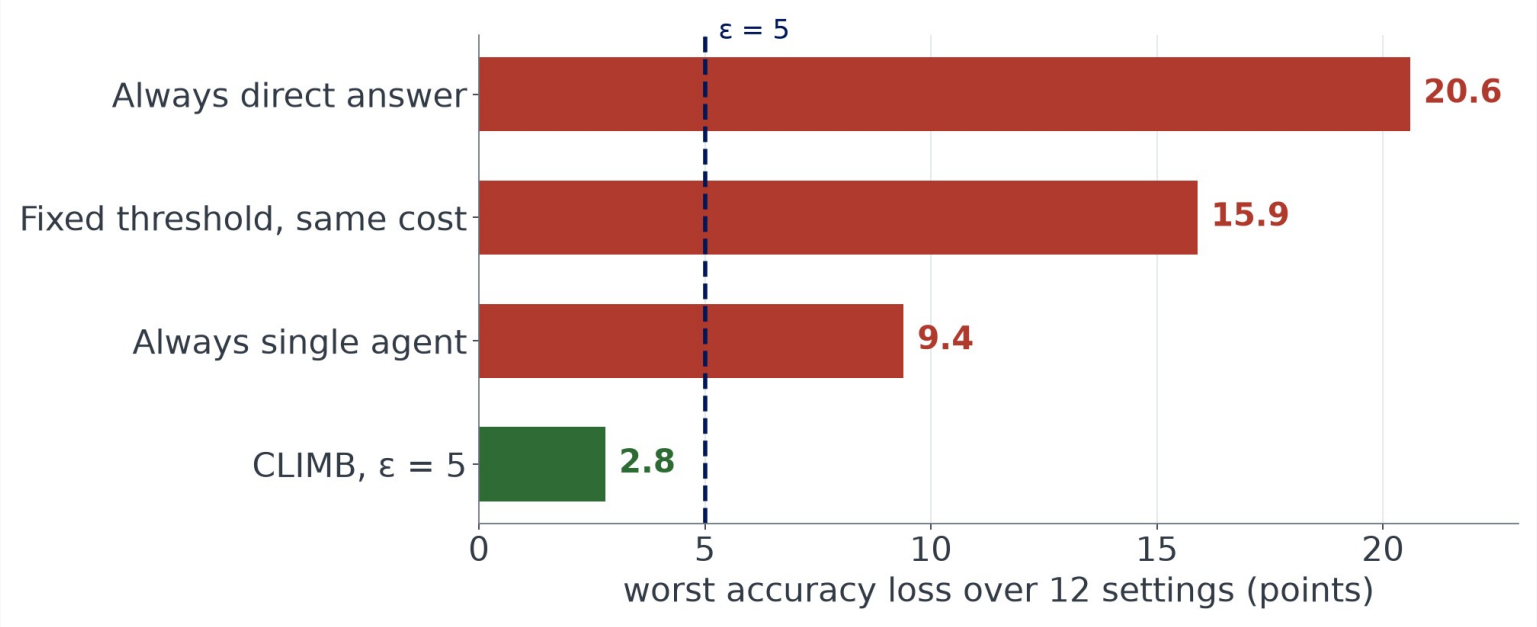
Look one rung up or two?

→ **let the data choose**

No rule passes the check? → always call the team. Worst case: no savings, never lost accuracy.

HOW TO CLIMB · RULE

Where to draw the line? Guessing can go badly



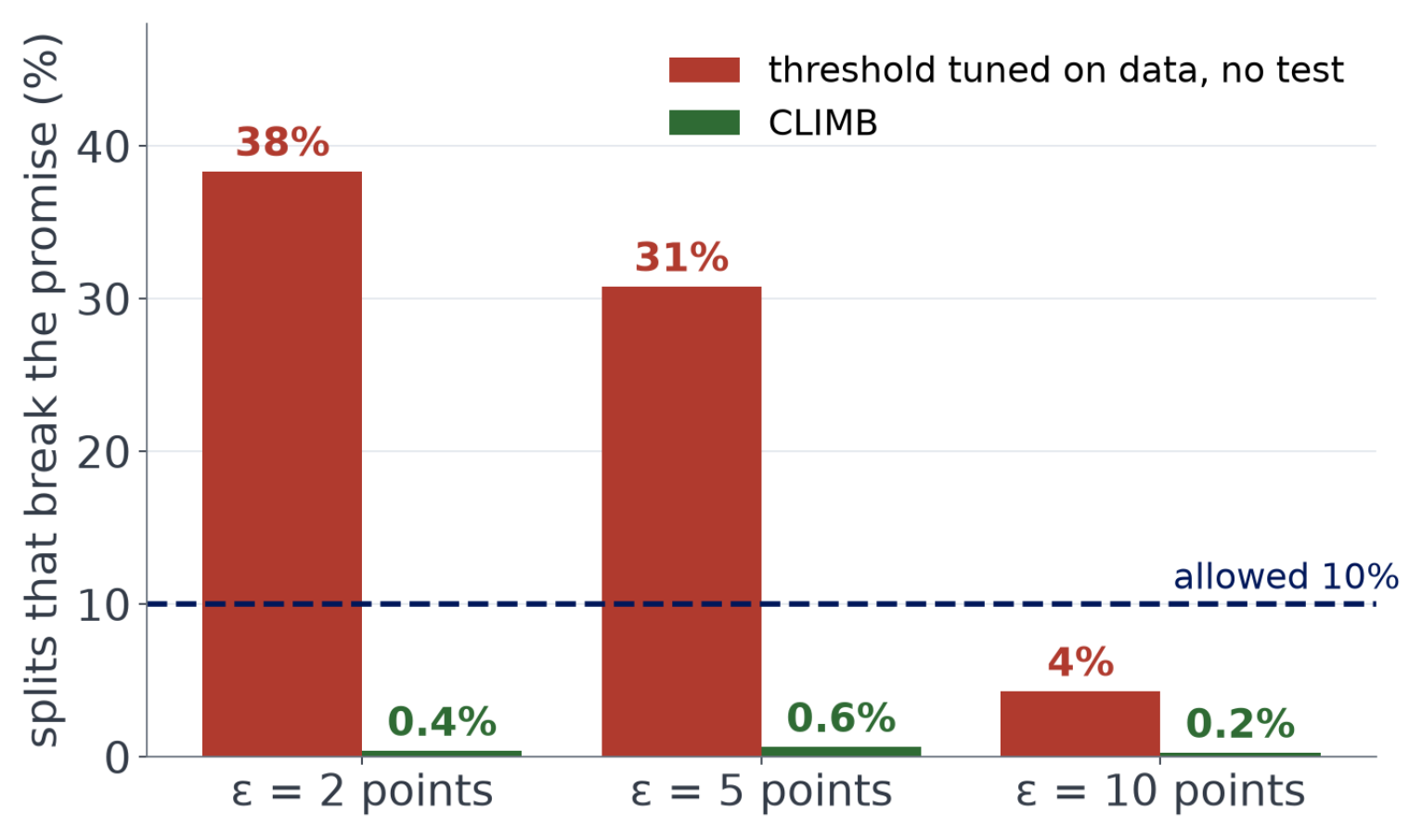
16 points

lost by a hand-picked threshold on one setting

Nothing at deployment says which setting will go wrong.

RESULTS

Without the test, the promise breaks



31%

of splits broken at ε = 5 by the tuned rule. CLIMB: 0.6%.

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

What a pass means, and what it does not

A bad rule reaches $20 at most 5% of the time.

So: reach $20 → accept the rule. This is wrong at most 5% of the time.

Right

"We accept it, and are wrong at most 5% of the time."

Wrong

"Reaching $20 proves the rule is good."

Wrong

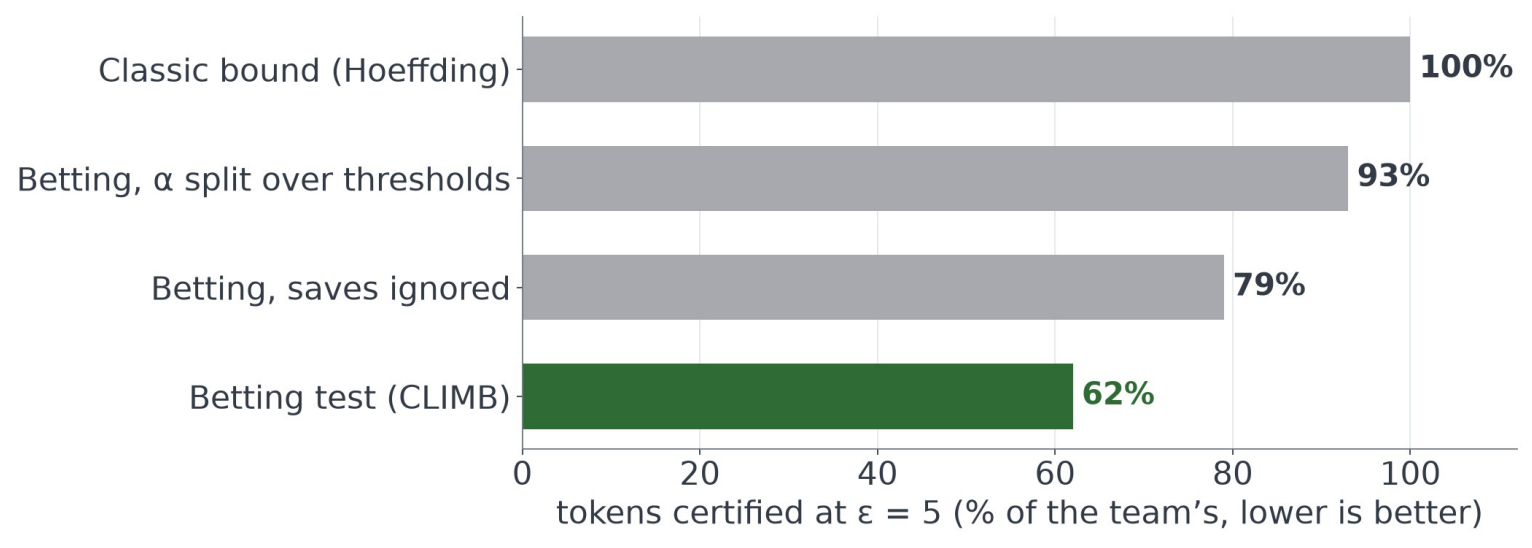
"5% of the rules that pass are bad."

Like a driving test. Bad drivers pass at most 5% of the time. Passing gets you a licence, it does not prove you can drive.

Two families, 5% each → the promise holds with probability ≥ 90%.

STEP 2 · SETTING THE THRESHOLD · THE BETTING TEST

Why bet on saved answers too? It saves more



62%

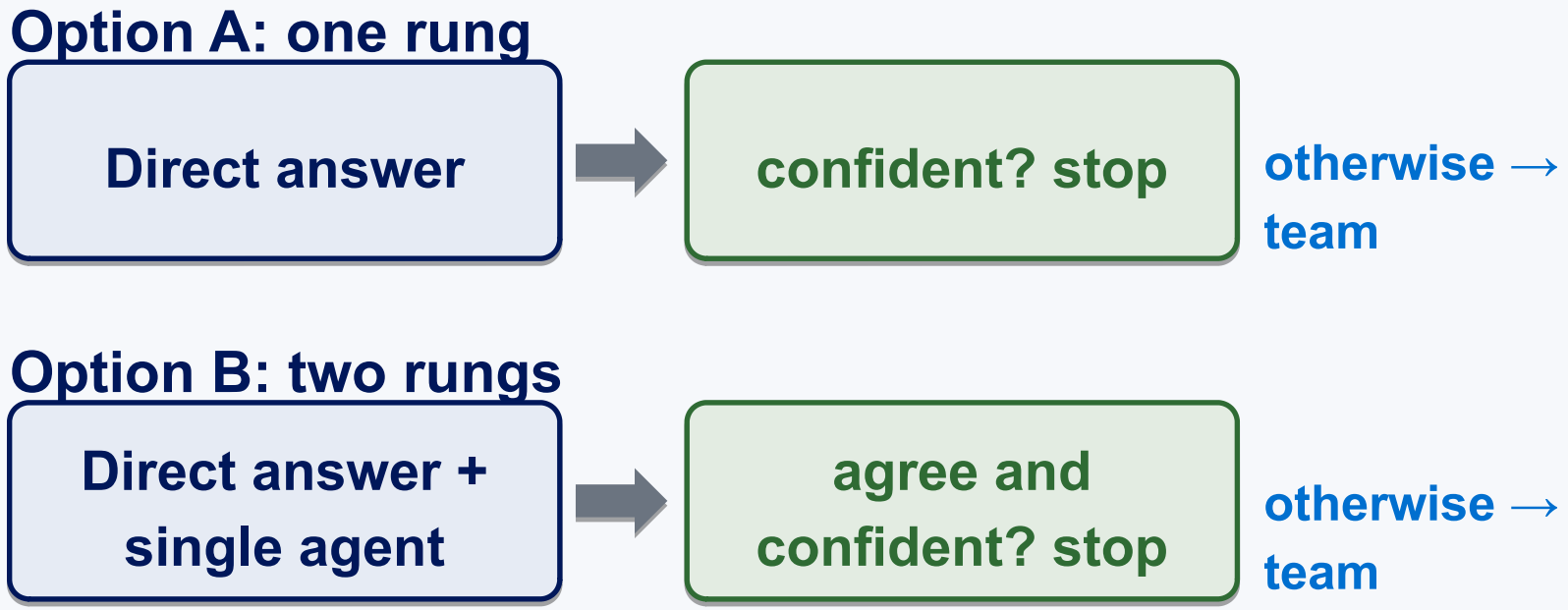
of the team's tokens when saves count. Losses only: 79%.

Teams break right answers on up to 9% of tasks. The bet gives the rule credit for them.

Tokens each valid test can certify at ε = 5. Lower is better.

HOW TO CLIMB · DEPTH

One rung or two? The data decides



Check both

on calibration data, then deploy the cheaper one that passes.

Neither passes? Always call the team.

EXPERIMENTS

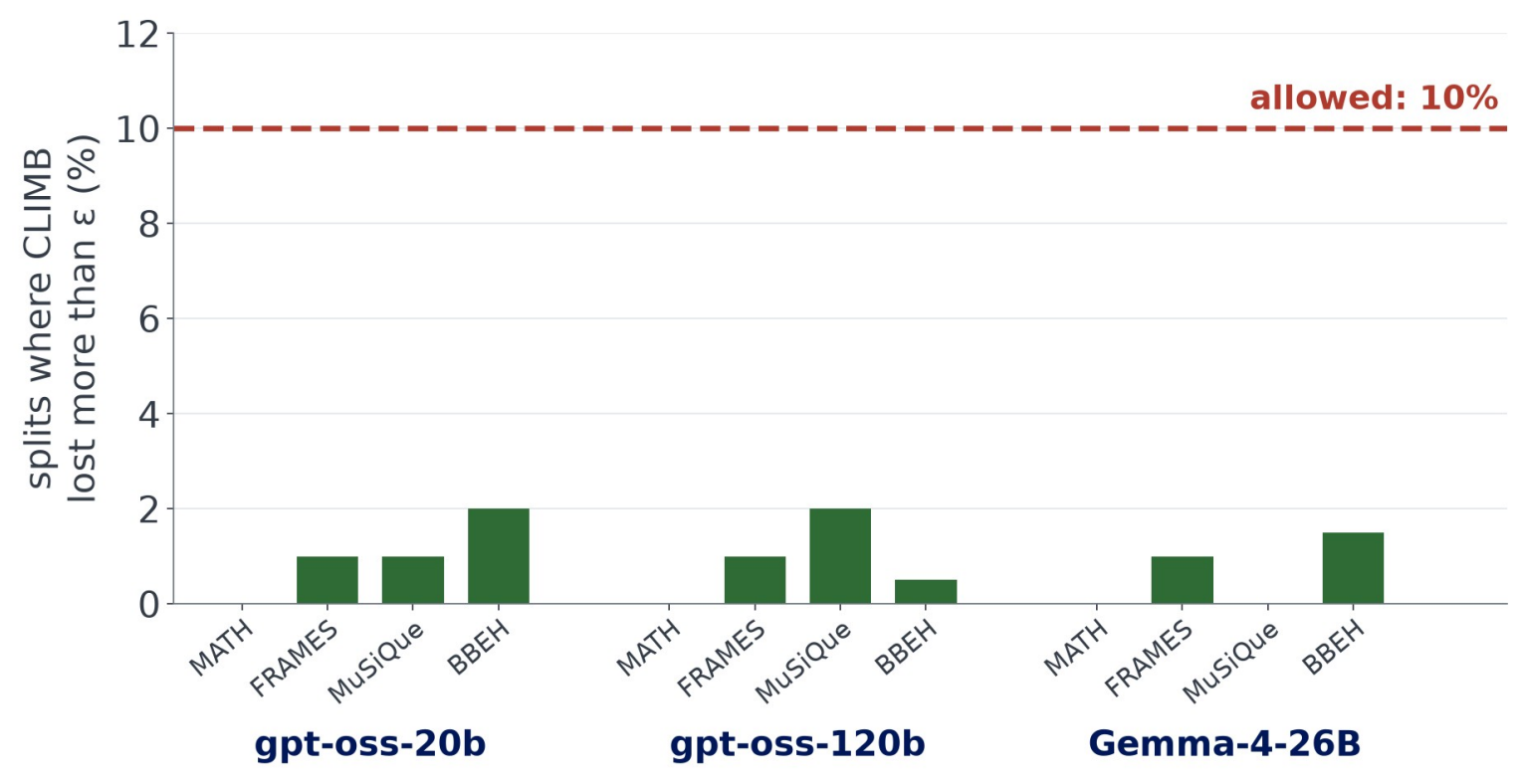
3 models × 4 benchmarks = 12 settings

	MATH math	FRAMES open-book QA	MuSiQue open-book QA	BBEH hard reasoning
gpt-oss-20b	**94.8%** team accuracy	**76.1%** team accuracy	**64.9%** team accuracy	**28.0%** team accuracy
gpt-oss-120b	**97.0%** team accuracy	**79.8%** team accuracy	**71.1%** team accuracy	**45.1%** team accuracy
Gemma-4-26B	**99.8%** team accuracy	**67.9%** team accuracy	**65.3%** team accuracy	**36.4%** team accuracy

Record every task once, then replay **200 random splits**: half to calibrate CLIMB, half to test it.

RESULTS

The promise holds on every setting

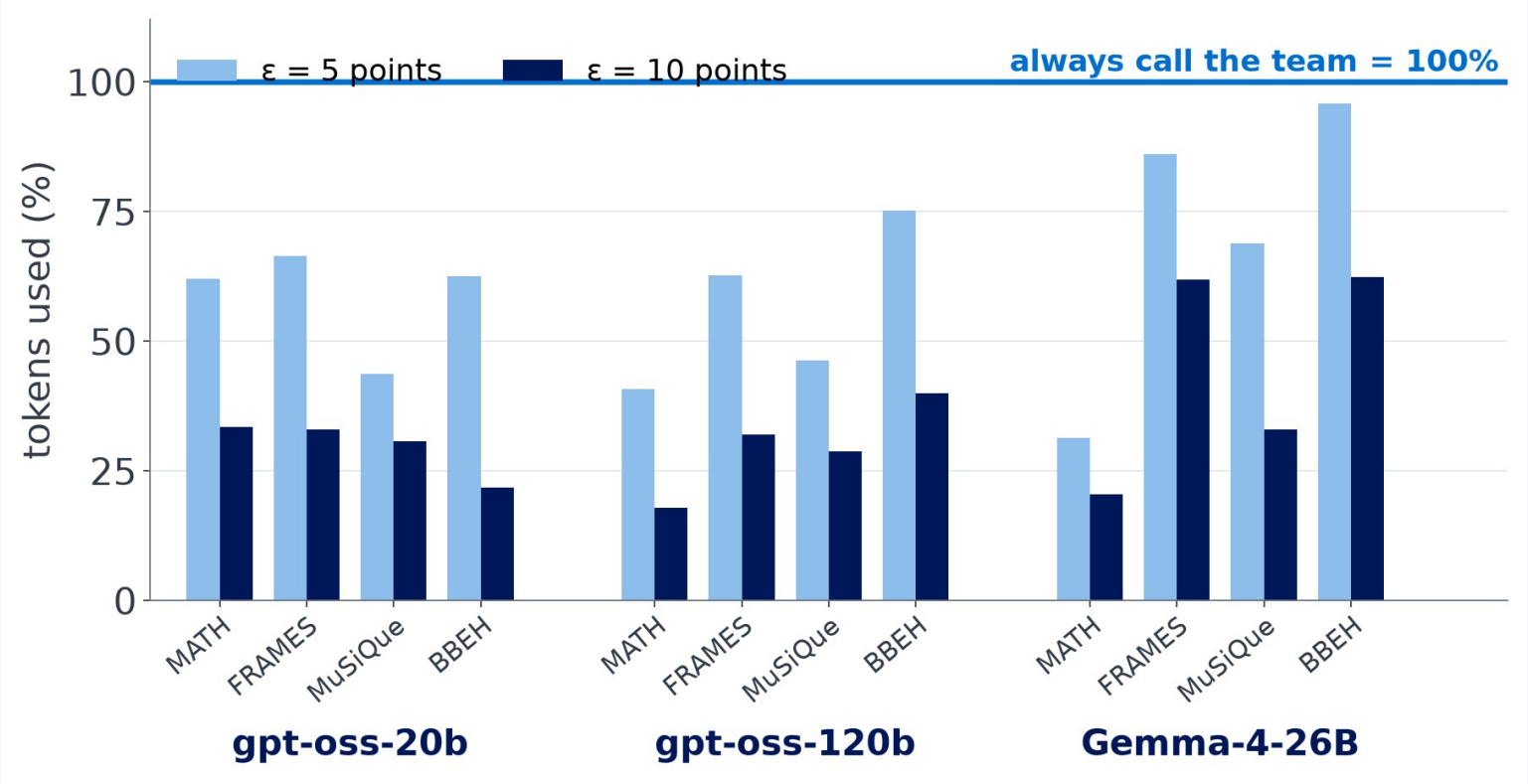


≤ 2%

of splits broke the promise. Allowed: 10%.

RESULTS

How much it saves



38% saved

allowing 5 points of loss

65% saved

allowing 10 points of loss